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Experiment-1 :- Use of Sequence Data Types in Python

Aim:- To study and implement different sequence data types in Python and perform operations on them. Theory:- Sequence data types are used to store multiple values in an ordered manner. They allow operations like indexing, slicing, iteration, and membership testing. Programs

a) Strings

```
s = "Hello Python"
print(s[0])
#Indexing
print(s[8:5])
# Slicing
print(s.lower())
# Operation
```

b) Lists

```
lst = [10, 20, 30, 40]
lst.append(50)
print(lst)
print(lst [1:3])
```

c) Arrays

```
import array as arr
a = arr.array('i', [1, 2, 3, 4])
a.append(5)
print(a)
```

d) Tuples

```
t = (1, 2, 3, 4)
print(t[0])
print(t[1:3])
```

f) Sets

```
s = {1, 2, 3, 3, 4}
print(s)
# duplicates removed
```

g) Range

```
r = range(1, 10, 2)
print(list(r))
```

Theory Questions & Answers

1.Sequence data types in Python What are sequence data types?

Sequence data types store multiple elements in an ordered manner, where each element has a specific position.

Types of sequence data types: List Tuple String Range Indexing Used to access elements based on their position. Starts from 0 for the first element. Negative indexing starts from the end (e.g., -1 is the last element). Slicing Used to extract a portion of the sequence. Defined by start, stop, and step. Works the same across lists, tuples, and strings. The start index is included, but the stop index is excluded.

2.Difference between List and Tuple

List Mutable (can be changed after creation) Uses more memory Slightly slower Suitable when data needs modification Tuple Immutable (cannot be changed after creation) Uses less memory Faster than lists Suitable for fixed data Why prefer tuple in real-world applications? Ensures data safety (cannot be modified accidentally) Better performance Can be used as dictionary keys (lists cannot) Useful for fixed collections like coordinates or database records

3.Arrays vs Lists

Lists Can store elements of different data types Flexible and widely used Arrays (array module) Store elements of same data type only More memory-efficient Faster for numerical operations When to use arrays instead of lists? When working with large amounts of numeric data When memory efficiency is important When all elements are of the same type

4.How dictionaries work internally

Dictionaries are based on a hash table. Each key is processed using a hash function to generate a unique index. The value is stored at that index. Why dictionary lookups are faster? They use direct access via hashing. Time complexity is $O(1)$ (constant time) in most cases. Lists and tuples require scanning elements, which takes $O(n)$ time. Collision handling: If two keys map to the same index, Python resolves it using internal techniques like probing.

5. Difference between Sets and Lists?

List: Ordered collection. Allows duplicate values. Supports indexing.
Set: Unordered collection. Does not allow duplicates. No indexing.
How sets handle duplicates: Duplicate values are automatically removed. Only unique elements are stored.
Where sets are useful: Removing duplicates from data, Fast membership testing, Performing mathematical operations like union, intersection, and difference.

Conclusion:- In this assignment, we learned about different sequence data types in Python such as strings, lists, arrays, tuples, dictionaries, sets, and range. We also performed operations like indexing and slicing, and understood their real-world applications.